Dictionary-Based Slang Normalization for Indonesian Sentiment Classification: A Controlled Comparison

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***Abstract*—We evaluate a fixed dictionary-based slang normalization policy on two Indonesian sentiment datasets using separately trained word and character TF-IDF support vector machines and IndoBERT. Shared audited splits, paired uncertainty estimates, direct representation contrasts, and fixed-model input switches distinguish complete pipeline comparisons from inference-only substitution. On SmSA, the normalized-minus-original IndoBERT pipeline effect averaged +0.0202 macro-F1 across three seeds, with a nominal 95% paired group-bootstrap interval of [0.0037, 0.0381]. Its lower bound did not exceed the investigator-selected +0.01 practical-gain threshold. The other five pipeline intervals and all four direct-contrast intervals included zero. Fixed-model substitution both corrected and introduced errors. These results characterize the tested datasets, models, and dictionary policy rather than establishing a general normalization rule.**

***Index Terms*—Indonesian sentiment analysis, slang normalization, text preprocessing, IndoBERT, character n-grams**

I. Introduction

Indonesian customer reviews and online comments contain abbreviations, spelling variants, and morphological transformations that differ from standard written forms [1], [2]. Sentiment classifiers can receive this wording directly or replace known informal forms with dictionary alternatives. Substitution may reduce variation, but ambiguous mappings and context-dependent meanings can also change the input inappropriately.

Prior work examines normalization with Indonesian pretrained models [3], [4] and compares broader preprocessing pipelines across representations [5]. Our contribution is the combined design: one isolated dictionary intervention, shared audited splits for two public datasets, separate word and character TF-IDF models alongside IndoBERT, and both retrained-pipeline and fixed-model input comparisons. None of these ingredients alone constitutes a new model or a first normalization experiment.

Three questions guide the study. *RQ1:* Does normalization benefit the word model more than the character model or IndoBERT? *RQ2:* Are the effects consistent across PRDECT-ID and SmSA? *RQ3:* Which predictions change when the same trained classifier receives normalized rather than original input? Direct contrasts address RQ1; dataset-specific effects address RQ2; and paired input-switch transitions address RQ3. Absolute scores across datasets with different label schemes are not treated as a controlled comparison of dataset difficulty. Conclusions concern the tested resources and configurations, not cross-domain transfer or semantic preservation.

II. Related Work

The Colloquial Indonesian Lexicon supplies informal–formal mappings, including ambiguous entries [1]; IndoCollex instead evaluates normalization as a word-transformation task [2]. Dictionary coverage or normalization accuracy alone does not establish downstream sentiment benefit. Babanejad et al. [6] distinguish preprocessing at representation-learning and downstream stages, motivating our explicit separation of classifier training and evaluation inputs.

IndoNLU provides SmSA and IndoBERT [7]; IndoBERTweet adapts an Indonesian encoder and vocabulary to Twitter [8]. Siva et al. [9] compare IndoBERT with an SVM combining word and character TF-IDF. Our comparison separates those TF-IDF representations and estimates their normalization effects on shared splits. PRDECT-ID [10] and resource collections such as NusaCrowd [11] support evaluation without new scraping or annotation.

Table I summarizes the nearest preprocessing studies. Bustamin et al. [3] combine dictionary lookup with edit distance. Budianoor et al. [4] isolate a slang variant relative to a cleaning baseline, while Prichanti et al. [5] compare bundled pipelines across representations. Kuswanto and Tanamal [12] report uncertainty for model comparisons and a preprocessing ablation for emotion classification. Our study combines an isolated substitution, separate representations, paired effects, and both fixed-model input directions; it does not claim that any one element is unprecedented.

III. Methodology

Fig. 1 presents the experimental workflow for evaluating the effect of slang normalization on Indonesian sentiment classification.

A. Datasets and Partitioning

PRDECT-ID contains 5,400 Indonesian Tokopedia product reviews across 29 categories, labeled positive or negative [10]. SmSA contains online comments and reviews with positive, negative, and neutral labels; its released train, validation, and

TABLE I
CLOSEST PRIOR STUDIES AND THE PRESENT DESIGN

Study	Data	Intervention and models	Replication / uncertainty	Fixed-input switch
Bustamin et al. [3]	Tweets	Dictionary and edit distance; IndoBERT	Three training/testing runs	–
Budianoor et al. [4]	Culinary comments	Slang versus cleaned baseline; IndoBERT	Seed 42	–
Prichanti et al. [5]	App reviews	Bundled processing; Word2Vec/logistic and IndoRoBERTa	Seed 42	–
Kuswanto and Tanamal [12]	App reviews	Bundled ablation for emotion; IndoBERT variants	Three ablation seeds; paired intervals for model comparisons	–
This study	Two review sets	Isolated substitution; word, character and IndoBERT	Paired group intervals; three IndoBERT seeds	Both directions

Dashes indicate that a crossed fixed-model input comparison was not reported in the checked study; they do not imply that related analyses are impossible.

test sets contain 11,000, 1,260, and 500 examples, respectively [7]. These are release sizes, not final sample counts. Only review text was used as input; ratings, emotion labels, and product metadata were excluded. SmSA's baseline uses its released, already preprocessed text without additional slang normalization.

The audit checked missing text and invalid labels, then detected duplicates using Unicode NFC normalization, case folding, and whitespace collapsing. Identical canonical texts with agreeing labels retained one representative; conflicting-label groups would be excluded. SmSA duplicates were resolved by partition priority (test, validation, training), then source order; PRDECT-ID used source order. Distinct texts that became identical only after normalization were retained, including any differing labels, because such collisions may reveal information lost through substitution.

Records sharing any canonical original or normalized string formed connected groups within each dataset. PRDECT-ID groups were assigned to approximate 70/15/15 partitions. The deterministic algorithm orders groups by decreasing size, breaks ties using SHA-256 of seed 42 and group identifier, and greedily minimizes squared class-count deficits divided by target counts, plus a total-size term weighted by 0.25. SmSA retained official assignments, with a rule to remove lower-priority members if a group crossed partitions. The dictionary, grouping implementation, tokenizer, and split assignments are jointly fingerprinted. Any change requires a new overlap audit and, if affected, rebuilt splits before training.

The audit removed 112 canonical duplicates from PRDECT-ID and 81 from SmSA's training partition; no missing/invalid records or conflicting-original-label groups were found. Table II gives retained counts; Supplementary Table S4 reports class support by split. SmSA validation and test membership remained unchanged. No canonical original or normalized text crosses partitions. Normalization changes 489 of 793 PRDECT-ID test reviews and 97 of 500 SmSA test reviews; the latter comprise 52 negative, 30 positive, and 15 neutral examples. These subgroup sizes limit the precision of class-specific conclusions.

TABLE II
AUDITED DATASET PARTITIONS

Dataset	Train	Validation	Test
PRDECT-ID	3,702	793	793
SmSA	10,919	1,260	500

B. Preprocessing and Dictionary Normalization

Let $O(x)$ denote minimal cleanup: Unicode NFC normalization and trimming and collapsing whitespace. Let $N(x)$ apply dictionary substitution once to $O(x)$. Both conditions retain punctuation, emojis, and negation. Neither includes stemming, stopword removal, edit-distance correction, or other spelling transformations.

The normalizer uses a pinned version of Kamus Alay, associated with the Colloquial Indonesian Lexicon [1]. Dictionary keys and values undergo NFC normalization, case folding, and whitespace cleanup. Eligible keys are single tokens consisting of Unicode letters or decimal digits; replacement values may contain multiple words. Empty replacements and keys with multiple distinct formal alternatives are excluded. Identity mappings do not count as changes. Matching is case-insensitive on complete, maximal letter-or-digit tokens, leaving unmatched text and surrounding symbols unchanged. Replacements use the stored formal value and are not processed again within the same pass.

The pinned dictionary snapshot contains 15,006 rows. Restricting keys to Unicode letters and decimal digits yields 4,202 distinct eligible keys; 71 with conflicting replacements were skipped, leaving 4,131 active mappings. Replacement logs record tokens, replacements, and positions. Four normalized collision groups remain in PRDECT-ID and one in SmSA, none with conflicting labels. Dictionary coverage is an operational measure, not a verified annotation of slang. The policy was checked using constructed examples without manual inspection of test reviews and was fixed before evaluation.

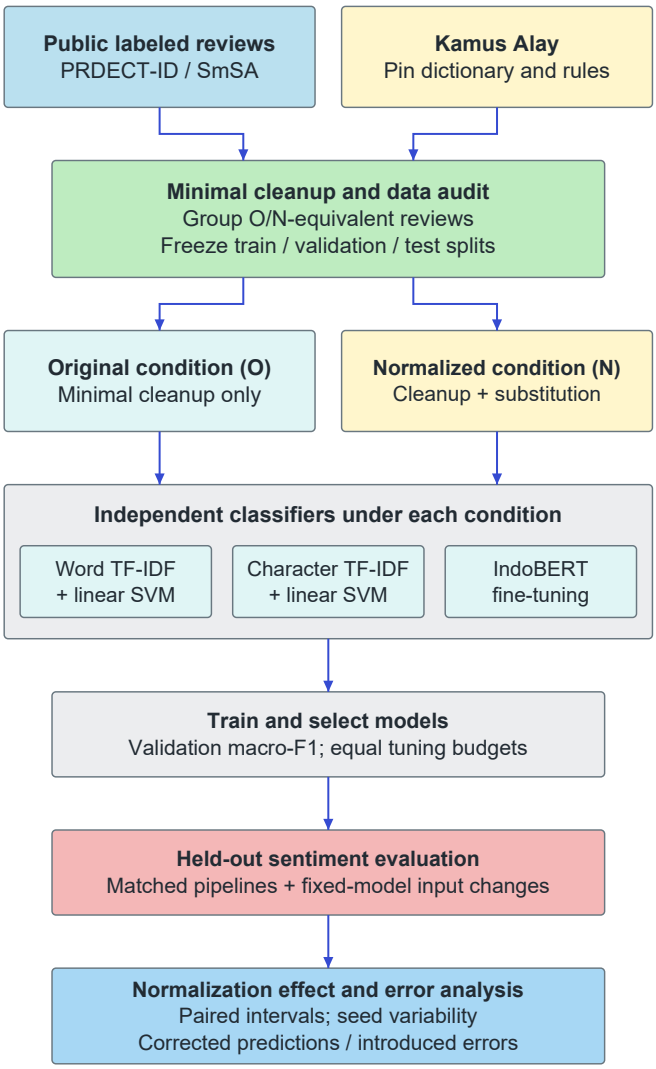


Fig. 1. Workflow for evaluating Indonesian slang normalization. O denotes minimal cleanup and N adds dictionary substitution; each classifier is trained separately on each branch.

On the held-out sets it made 1,527 replacements in PRDECT-ID and 121 in SmSA; counts are operational matches, not verified slang prevalence.

C. Models and Training

The two datasets, three classifiers, and two preprocessing conditions formed 12 configurations; both conditions used the same retained examples and split assignments. The first two classifiers use word and character TF-IDF with a linear support vector machine (SVM). Word features include unigrams and bigrams over maximal letter-or-digit sequences and individual remaining non-whitespace symbols; character features span three through five characters, including spaces and punctuation. Both analyzers case-fold their inputs. Vocabulary and inverse document frequencies were fitted separately on training data under each condition, with minimum document frequency two, sublinear term frequency, smoothed inverse document frequency, and L_2 normalization. Limits were 50,000 word and 100,000 character features.

TABLE III
TUNING AND TRAINING SETTINGS

Component	Setting
Linear SVM	$C \in \{0.1, 1, 10\}$; unweighted classes; squared hinge loss with L_2 penalty
IndoBERT search	Learning rate $\in \{2, 3, 5\} \times 10^{-5}$; tuning seed 42
Optimization	AdamW; weight decay 0.01; effective batch size 16; gradient clipping at 1.0
Schedule	Five epochs; 10% linear warmup, followed by linear decay
Selection	Highest validation macro-F1; evaluation after each epoch for IndoBERT
Final runs	SVM: seed 42; IndoBERT: seeds 13, 37, 73

The contextual classifier fine-tuned the pretrained `indobenchmark/indobert-base-p1` checkpoint [7], with a linear head on its pooled representation and unweighted cross-entropy updating all layers. A serialized lowercase BERT WordPiece tokenizer uses the pinned vocabulary. Selecting the smallest length in $\{128, 256, 512\}$ covering at least 95% of training examples under each condition, including special tokens, yielded 128 for both datasets. Training truncation affects 14 original and 13 normalized PRDECT-ID reviews, and one review per condition in SmSA. All 489 and 97 text-changed test reviews in PRDECT-ID and SmSA, respectively, also produced different word-token sequences and truncated BERT token IDs. A post hoc vector-survival diagnostic found that, among text-changed test reviews, word TF-IDF vectors changed for 488/489 under the original-fitted PRDECT-ID vocabulary and 489/489 under the normalized-fitted vocabulary; vectors changed for 95/97 under either fixed SmSA word vectorizer. Under either fixed character vectorizer, they changed for 489/489 and 97/97, respectively. These counts show any vector change, not the fate of each replacement. A pretrained base checkpoint avoids unknown sentiment-fine-tuning exposure, although pretraining overlap cannot be ruled out.

Table III defines equal search budgets across preprocessing conditions: three SVM settings or three transformer tuning runs per model–dataset–condition. SVM training used tolerance 10^{-4} and a 20,000-iteration limit; convergence was checked before evaluation. For IndoBERT, bias and layer-normalization parameters were exempt from weight decay, pretrained dropout settings were retained, and batch size was 16. Validation ties favored smaller C, or earlier epochs followed by lower learning rates. After learning-rate selection, IndoBERT was retrained from the pretrained checkpoint with three paired seeds, selecting each checkpoint by validation macro-F1. Validation data were not added to training, and the test set did not guide configuration changes. Selected SVM C

was 1 except for SmSA word-normalized ($C = 10$). Selected IndoBERT learning rates were 2×10^{-5} except for SmSA normalized (3×10^{-5}); Supplementary Table S5 reports final seed-specific epochs. These separately selected settings make the primary difference a full tuning-and-preprocessing pipeline comparison.

D. Evaluation and Uncertainty

The primary comparison uses separately tuned pipelines; fixed-model diagnostics hold the classifier constant while changing only its evaluation input. Macro-F1 is the primary metric, computed over the fixed label set of each dataset:

$$M = \frac{1}{K} \sum_{k=1}^{K} \frac{2\mathrm{TP}_k}{2\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k}, \tag{1}$$

where K is two for PRDECT-ID and three for SmSA; a zero denominator contributes zero. Accuracy and per-class performance provide supplementary evidence. For each dataset and classifier, the main effect is

$$\Delta M = M(f_N(N(X)), Y) - M(f_O(O(X)), Y), \tag{2}$$

where f_O and f_N are the classifiers trained under the respective conditions, and (X, Y) is the shared test set. Transformer scores and paired differences are summarized by their mean and sample standard deviation across seeds. Absolute scores are not pooled across datasets.

Following the emphasis on effect sizes and uncertainty in NLP evaluation [13], 95% percentile bootstrap intervals accompany the differences. We draw 2,000 paired bootstrap samples of test groups with replacement and seed 42. The same samples are used across classifiers, conditions, and transformer seeds; each replicate averages the seed-specific transformer differences. Intervals describe test-sample uncertainty conditional on trained models; seed variability is reported separately.

To test whether normalization benefits the tested word model more than the alternatives, we predefine two contrasts per dataset:

$$D_{w,c} = \Delta M_{\mathrm{word}} - \Delta M_{\mathrm{char}}, \tag{3}$$

$$D_{w,b} = \Delta M_{\mathrm{word}} - \overline{\Delta M}_{\mathrm{IndoBERT}}. \tag{4}$$

Each bootstrap replicate recomputes these contrasts on identical sampled reviews. The six pipeline effects and four direct contrasts use nominal intervals, not simultaneous confidence guarantees. A positive effect for one model and an uncertain effect for another does not alone establish a difference between them.

Before classification experiments, we selected 0.01 macro-F1 (one percentage point) as a practical reporting reference. No further rationale was recorded; it is not a community standard. An interval wholly above 0.01 supports a practically meaningful gain within the tested setting; intervals crossing zero or the margin require caution. An interval containing zero does not establish equivalence, and this study is not a powered equivalence test. Per-class harms and introduced errors accompany aggregate interpretation.

TABLE IV
HELD-OUT MACRO-F1 FOR ALL TRAINING/INPUT CONDITIONS

Data	Model	O/O	O/N	N/O	N/N
PRDECT	Word	.9570	.9583	.9369	.9646
	Char.	.9595	.9646	.9495	.9608
	BERT	.9844	.9836	.9810	.9810
SmSA	Word	.7350	.7366	.7358	.7414
	Char.	.7214	.7244	.7342	.7317
	BERT	.8830	.8836	.9055	.9032

First symbol: training condition; second: evaluation input. BERT cells average three paired seeds on the same reviews.

E. Input Diagnostics and Error Analysis

Each selected classifier also predicts both $O(X)$ and $N(X)$, producing the four training/evaluation combinations in Table IV, without additional training or test-based tuning. For each fixed model, prediction transitions count incorrect-to-correct, correct-to-incorrect, correct-to-correct, and incorrect-to-incorrect cases. This distinguishes input-induced changes from the effects of retraining.

Changed and unchanged reviews are defined by $N(x) \neq O(x)$ before inspecting predictions. Identical inputs must yield identical predictions for a fixed model; this also serves as an implementation check. Any qualitative inspection of corrected and introduced errors is descriptive and does not create new gold labels or estimate linguistic error prevalence. Sentiment improvements alone do not establish semantic preservation.

F. Reproducibility

The full local workspace retains source checksums, split identifiers, selection records, and per-review predictions. The local snapshot `review_snapshot_v2` includes manuscript and supplement sources, preview PDFs, code, configurations, manifests, and aggregate results; it excludes review-level data and predictions, dictionary mappings, weights, private payloads, and credentials. It is not a public or reviewer-accessible release. The local `REPRODUCIBILITY.md` guide documents retrieval, auditing, training, inference, analysis, and figure steps; exact replay additionally requires privately retained artifacts, while fresh retraining need not reproduce the same weights. SVM training used Python 3.12.14 and scikit-learn 1.7.1; IndoBERT used Python 3.13.15, PyTorch 2.11.0, Transformers 5.16.1, and one Tesla T4. The cloud image and package inventory were recorded. Resource terms and release status are summarized in the supplement.

IV. RESULTS

Table IV shows all four training/input cells, and Table V reports the matched-pipeline effects and intervals. On PRDECT-ID, the word SVM gained 0.0076 macro-F1, the character SVM gained 0.0013, and mean IndoBERT fell by 0.0034; all three nominal intervals include zero. On SmSA, the word and character SVM effects were 0.0064 and 0.0103, respectively, with intervals spanning zero. Mean IndoBERT improved by 0.0202, with an interval of [0.0037, 0.0381]. Although this

TABLE V
HELD-OUT MATCHED-PIPELINE MACRO-F1 AND NORMALIZATION EFFECTS

Data	Model	O/O	N/N	Δ	95% CI
PRDECT	Word	.9570	.9646	+.0076	[−.0013, +.0177]
	Char.	.9595	.9608	+.0013	[−.0076, +.0101]
	BERT	.9844	.9810	−.0034	[−.0097, +.0025]
SmSA	Word	.7350	.7414	+.0064	[−.0135, +.0297]
	Char.	.7214	.7317	+.0103	[−.0046, +.0267]
	BERT	.8830	.9032	+.0202	[+.0037, +.0381]

O/O and N/N denote matched training/evaluation conditions. BERT values are means over three seeds. Intervals are nominal, paired, and conditional on the fitted models.

TABLE VI
DIRECT CONTRASTS BETWEEN NORMALIZATION EFFECTS

Data	Contrast	Estimate	95% CI
PRDECT	Word − character	+.0063	[−.0038, +.0178]
	Word − IndoBERT	+.0110	[−.0004, +.0233]
SmSA	Word − character	−.0039	[−.0303, +.0226]
	Word − IndoBERT	−.0138	[−.0384, +.0105]

Intervals use the same paired group draws as the pipeline effects.

interval is above zero, its lower bound is below the pre-experiment practical threshold of 0.01. Thus none of the six intervals supports a gain exceeding that threshold under the specified rule. Fig. 2 displays these estimates and intervals.

The paired standard deviations of the three IndoBERT effects were 0.0090 for PRDECT-ID and 0.0088 for SmSA. Supplementary Table S1 lists the paired score and difference for each seed on the same held-out reviews; the seeds do not create independent test samples. These standard deviations describe seed variation, whereas the intervals describe resampling uncertainty conditional on the fitted models. They are not interchangeable, and the nominal intervals are not simultaneous guarantees across the six effects and four contrasts.

A. Direct Representation Contrasts

Table VI reports the four predefined differences between normalization effects. All four intervals include zero. In particular, the apparently larger SmSA IndoBERT effect does not by itself establish that its normalization effect differs from the word SVM's effect. These intervals also do not establish equivalence between representations.

B. Class and Fixed-Model Diagnostics

Table VII summarizes class F1 with unique-review support; Supplementary Table S2 gives four-decimal values. On SmSA, the word SVM's neutral F1 increased while its negative and positive F1 declined. IndoBERT's mean neutral F1 also increased. These class results are descriptive, not separately adjusted tests.

Table VIII gives both fixed-model input directions. For an original-trained model, the reported change is $O/N - O/O$; for a normalized-trained model, it is $N/O - N/N$. Counts describe transitions from O/O to O/N and from N/N to N/O, respectively. IndoBERT entries are means across the three seeds on the same reviews, not unique-review totals; Supplementary Table S3 also reports distinct-review unions.

For SmSA IndoBERT, the matched-pipeline change was $+.0202$, whereas applying N to the original-trained model changed macro-F1 by $+.0006$ and changed one prediction per seed on average. These are different comparisons. Their difference does not causally separate substitution, retraining, or independently selected hyperparameters.

V. DISCUSSION AND LIMITATIONS

The research questions receive different levels of resolution. For RQ1, all direct-contrast intervals include zero, leaving representation differences unresolved rather than establishing equivalence. For RQ2, point estimates differ descriptively by dataset: the SmSA IndoBERT interval is positive, while the PRDECT-ID IndoBERT interval crosses zero. The PRDECT-ID word interval spans a small negative value through a gain above 0.01, and the SmSA IndoBERT interval spans positive effects both below and above that margin. These ranges leave materially different outcomes unresolved. For RQ3, fixed-model input switches corrected and introduced errors in both directions (Table VIII).

For practice, the relevant decision is to evaluate a complete preprocessing and training configuration on data representative of its intended use. Adding substitution only at inference is a separate decision, because the fixed-model results need not equal a separately tuned pipeline difference. Potential explanations involving feature sharing, subword robustness, or training adaptation are post hoc hypotheses, not mechanisms established by these comparisons. In particular, the observed SmSA IndoBERT pipeline and fixed-model differences do not identify separate causal contributions of substitution, retraining, or tuning.

The findings are limited to two datasets, one dictionary policy, and one pretrained transformer. The dictionary changed 489 of 793 PRDECT-ID test reviews and 97 of 500 SmSA reviews; only 15 changed SmSA reviews were neutral. Matches are not independently verified slang annotations, and sentiment scores do not establish semantic preservation. IndoBERT uses only three seeds; the nominal test-sample intervals are conditional on fitted models and do not provide simultaneous coverage. Pretraining overlap cannot be excluded, and normalization-only runtime was not measured.

The SVM selection was bound to local checkpoint hashes before test inference. The frozen IndoBERT selection recorded versioned checkpoint identifiers but not whole-weight hashes; all 12 cloud files were hashed subsequently and reproduced their saved validation predictions before held-out inference. This supports execution provenance without a pre-freeze cryptographic binding of the transformer weight bytes.

VI. CONCLUSION

A fixed dictionary intervention produced small and uncertain pipeline effects in most tested settings. SmSA IndoBERT showed a positive nominal effect, but its interval did not clear the predefined practical-gain threshold, and direct contrasts did

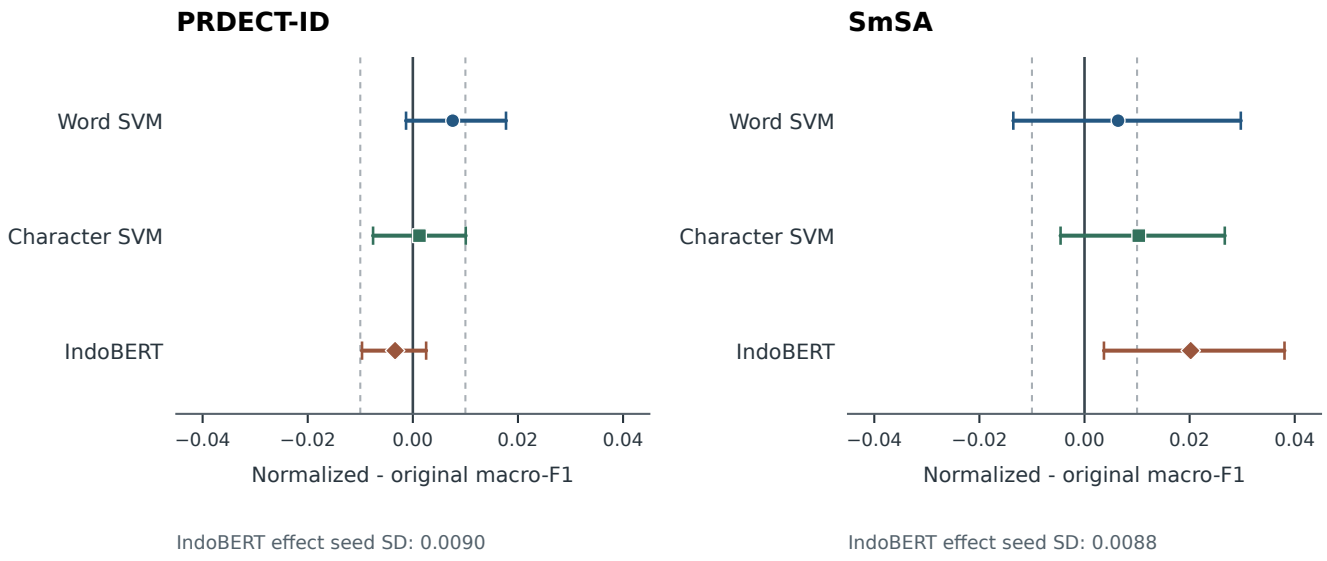


Fig. 2. Normalized-minus-original pipeline macro-F1 with nominal 95% paired group-bootstrap percentile intervals. Dashed lines mark the investigator-selected ± 0.01 practical margin; seed standard deviations describe variation across three IndoBERT effects separately.

TABLE VII
HELD-OUT CLASS F1 FOR MATCHED PIPELINES

Data	Class	Support	Word SVM		Character SVM		IndoBERT	
			O/O	N/N	O/O	N/N	O/O	N/N
PRDECT	Negative	412	.9593	.9666	.9618	.9631	.9852	.9820
	Positive	381	.9547	.9626	.9572	.9585	.9836	.9801
SmSA	Negative	204	.8426	.8337	.8145	.8238	.9372	.9495
	Neutral	88	.5263	.5674	.5507	.5612	.7769	.8248
	Positive	208	.8363	.8232	.7990	.8103	.9348	.9352

Supports count distinct test reviews; IndoBERT values are means of seed-specific F1, not pooled predictions.

TABLE VIII
FIXED-MODEL MACRO-F1 CHANGES AND PREDICTION TRANSITIONS

Data	Model	$O/N - O/O$	$N/O - N/N$
PRDECT	Word	+.0013 (10/9)	−.0277 (7/29)
	Char.	+.0051 (5/1)	−.0113 (8/17)
	BERT	−.0008 (.67/1.33)	.0000 (1.67/1.67)
SmSA	Word	+.0016 (1/0)	−.0056 (1/3)
	Char.	+.0030 (2/0)	+.0025 (1/1)
	BERT	+.0006 (.33/.33)	+.0023 (1.33/0)

Each entry is macro-F1 change (corrected/introduced errors). BERT counts are seed means, not unique reviews; changes are rounded to four decimals.

not resolve differences between representations. Fixed-model transitions further show that substitutions can both correct and introduce errors. Broader datasets, dictionary policies, and independently measured preprocessing costs are needed before extending these findings beyond the tested configurations.

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